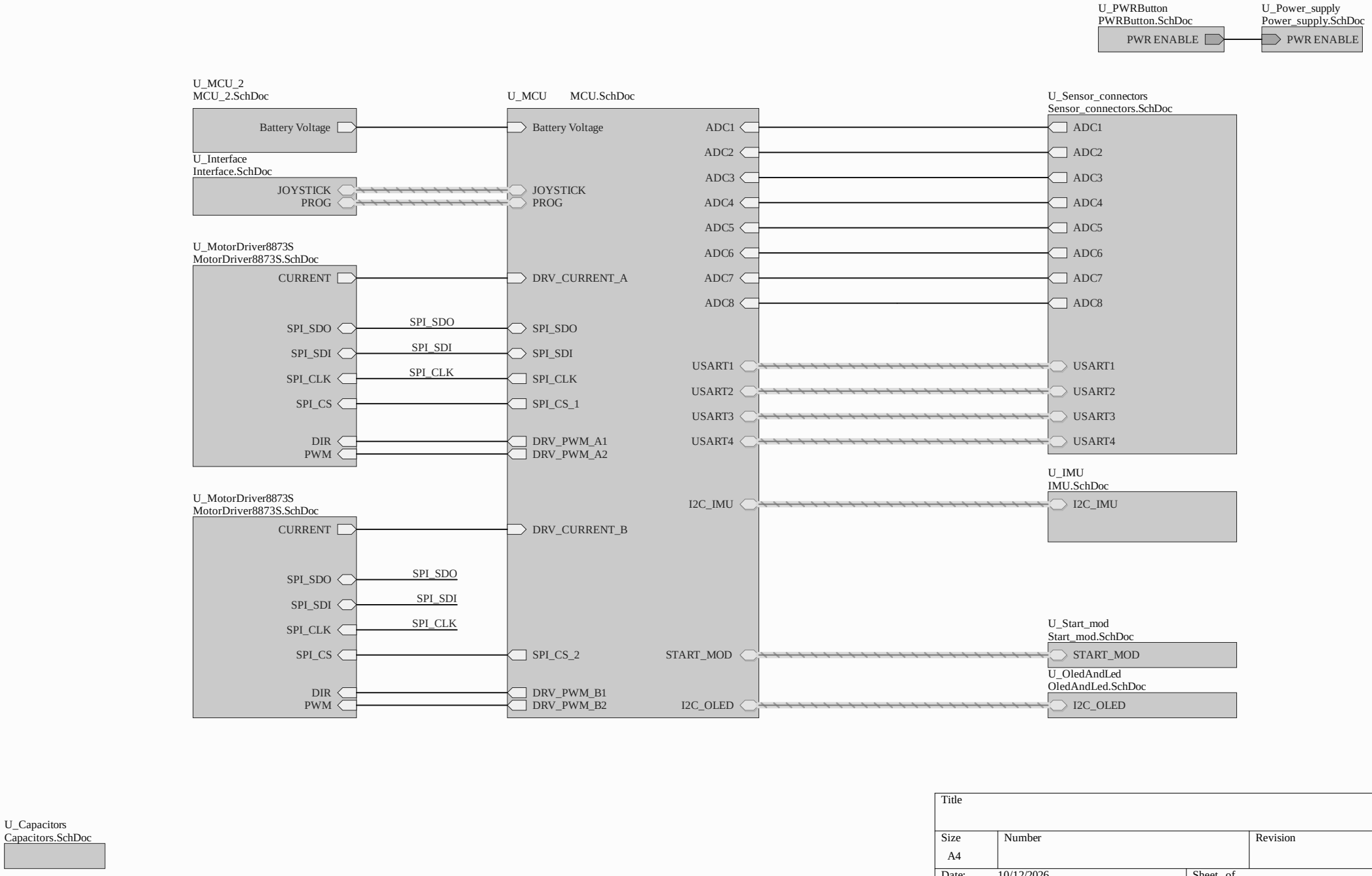
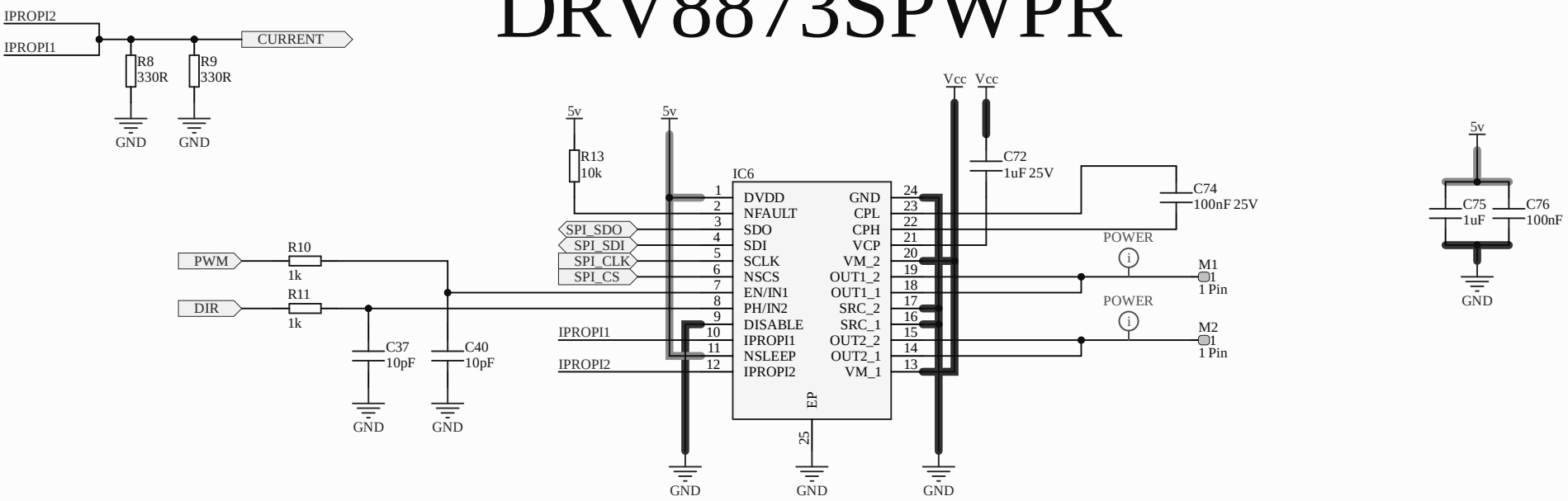

U_PWRButton
PWRButton.SchDoc
PWR ENABLE
U_Power_supply
Power_supply.SchDoc
PWR ENABLE
U_MCU_2
MCU_2.SchDoc
Battery Voltage
U_Interface
Interface.SchDoc
JOYSTICK
PROG
U_MotorDriver8873S
MotorDriver8873S.SchDoc
CURRENT
SPI_SDO
SPI_SDI
SPI_CLK
SPI_CS
DIR
PWM
U_MotorDriver8873S
MotorDriver8873S.SchDoc
CURRENT
SPI_SDO
SPI_SDI
SPI_CLK
SPI_CS
DIR
PWM
U_MCU
MCU.SchDoc
Battery Voltage
JOYSTICK
PROG
DRV_CURRENT_A
SPI_SDO
SPI_SDI
SPI_CLK
SPI_CS_1
DRV_PWM_A1
DRV_PWM_A2
DRV_CURRENT_B
SPI_CS_2
DRV_PWM_B1
DRV_PWM_B2
ADC1
ADC2
ADC3
ADC4
ADC5
ADC6
ADC7
ADC8
USART1
USART2
USART3
USART4
I2C_IMU
START_MOD
I2C_OLED
U_Sensor_connectors
Sensor_connectors.SchDoc
ADC1
ADC2
ADC3
ADC4
ADC5
ADC6
ADC7
ADC8
USART1
USART2
USART3
USART4
U_IMU
IMU.SchDoc
I2C_IMU
U_Start_mod
Start_mod.SchDoc
START_MOD
U_OledAndLed
OledAndLed.SchDoc
I2C_OLED
U_Capacitors
Capacitors.SchDoc
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Size
A4
Number
Revision
Date:
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Sheet of
File:
D:\tech\..\PCB.SchDoc
Drawn By:

DRV8873SPWPR



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File: D:\tech\..\MotorDriver8873S.SchDoc		Drawn By:

POWER FILTERING

Kazde Vdd/Vbat powinno miec 1uF + 100nF jak najblizej!!!

3.3v
C13 1uF
C18 100nF
GND

3.3v
C1 1uF
C3 100nF
GND

3.3v
C2 1uF
C4 100nF
GND

MCU things

CLK1
Y1
4 GND_2
1 XTAL_1
XTAL_2 3
GND_1 2
US3200005Z
CLK2
C21 6.8pF
C22 6.8pF
GND
GND

GLOBAL PORTS!

PROG
PROG
RESET — NRST
SWDIO — SWDIO
GND
SWCLK — SWCLK
GND

JOYSTICK
JOYSTICK
1 — JOY1
2 — JOY2
3 — JOY3
4 — JOY4
5 — JOY5

I2C
I2C_OLED
SCL — I2C_OLED_SCL
SDA — I2C_OLED_SDA

I2C
I2C_IMU
SCL — IMU_SCL
SDA — IMU_SDA

USART
USART1
TX — UART2_TX
RX — UART1_RX

USART
USART2
TX — UART2_TX
RX — UART2_RX

USART
USART3
TX — UART2_TX
RX — UART3_RX

USART
USART4
TX — UART2_TX
RX — UART4_RX

START_MOD
START_MOD
LED_RED — START_MOD_LED_1
LED_BLUE — START_MOD_LED_2
LED_GREEN — START_MOD_LED_3
IR — START_MOD_IR_1

IMU_SCL
IMU_SDA
START_MOD_LED_3
UART2_RX
UART2_TX
START_MOD_LED_1
UART4_RX
START_MOD_LED_2
SWCLK
3.3v
GND
3.3v
DRV_PWM_B1
DRV_PWM_B2

IC1 STM32G431CBU6
3.3v

EP
VDD_3
PB9
PB8-BOOT0
PB7
PB6
PB5
PB4
PB3
PC11
PC10
PA15
PA14

1 VBAT
JOY3 2 PC13
JOY2 3 PC14-OSC32_IN
JOY1 4 PC15-OSC32_OUT
CLK1 5 PF0-OSC_IN
CLK2 6 PF1-OSC_OUT
JOY5 NRST 7 PG10-NRST
DRV_CURRENT_B 8 PA0
ADC3 9 PA1
UART1_RX 10 PA2
SPI_CS_2 11 PA3
Battery Voltage 12 PA4

PA13 36 SWDIO
VDD_2 35
PA12 34 DRV_PWM_A2
PA11 33 DRV_PWM_A1
PA10 32 SPI_CS_1
PA9 31 I2C_OLED_SCL
PA8 30 I2C_OLED_SDA
PC6 29
PB15 28 SPI_SDI
PB14 27 SPI_SDO
PB13 26 SPI_CLK
PB12 25 DRV_CURRENT_A

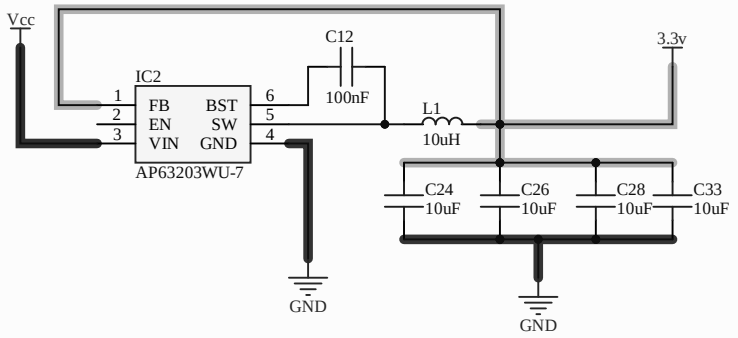
PA5
PA6
PA7
PC4
PB0
PB1
PB2
VREF+
VDDA
PB10
VDD_1
PB11

13 ADC4
14 ADC1
15 ADC2
16 ADC5
17 ADC6
18 ADC7
19 ADC8
20
21
22
23
24

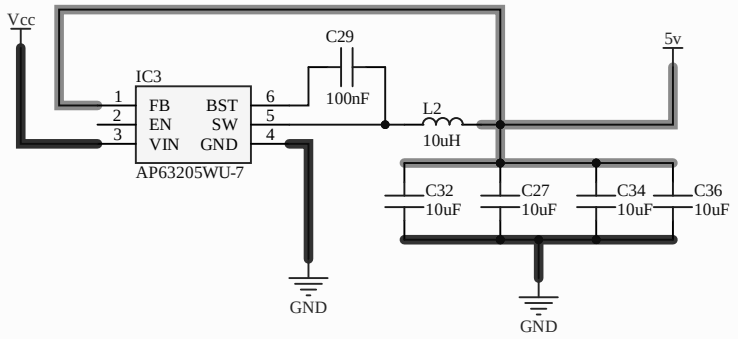
START_MOD_IR_1
UART3_RX

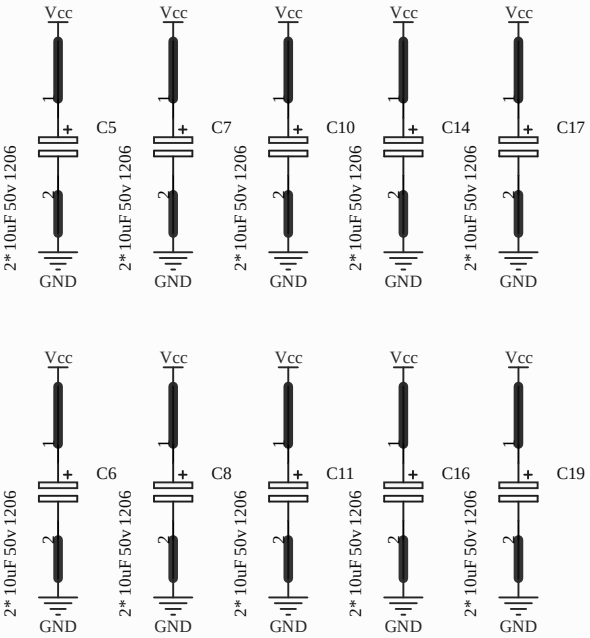
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Size A4	Number	Revision
Date: 10/12/2026		Sheet of
File: D:\tech\..\MCU.SchDoc		Drawn By:

Vcc to 3.3v pulse converter

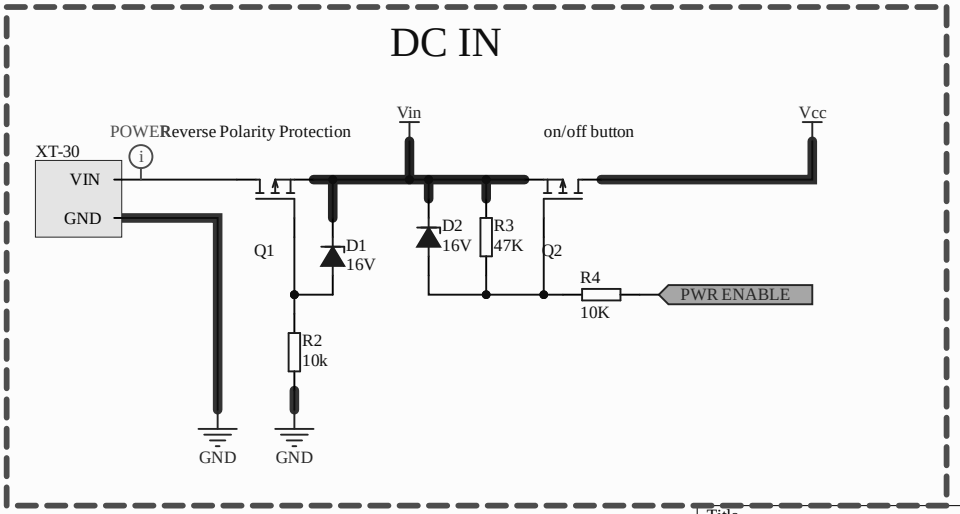


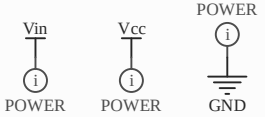
Vcc to 5v pulse converter





DC IN

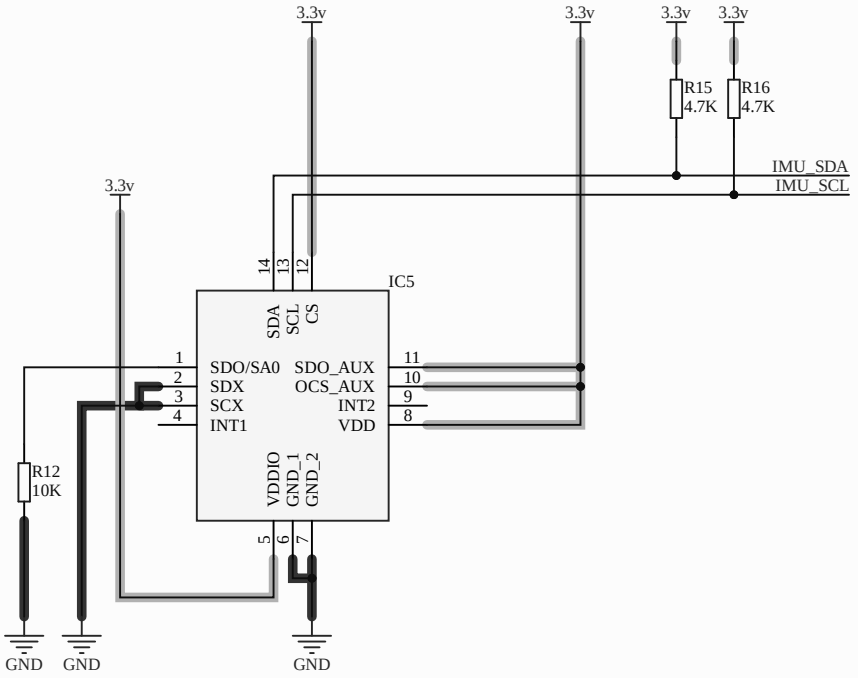


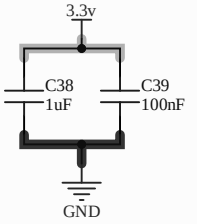


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IMU

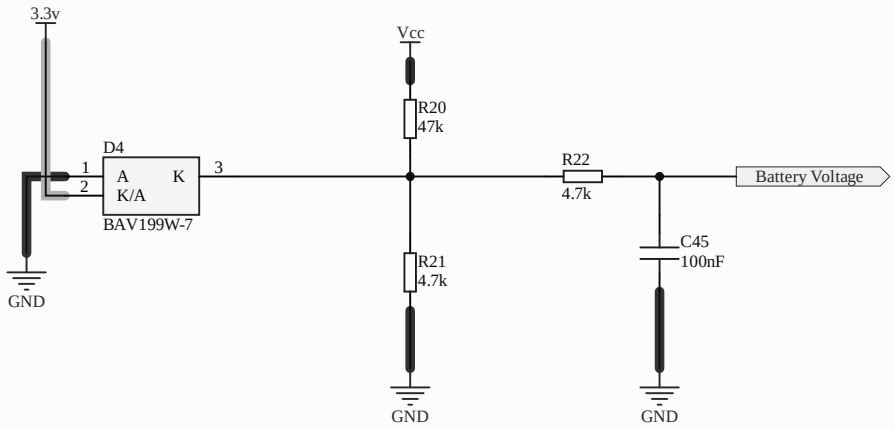


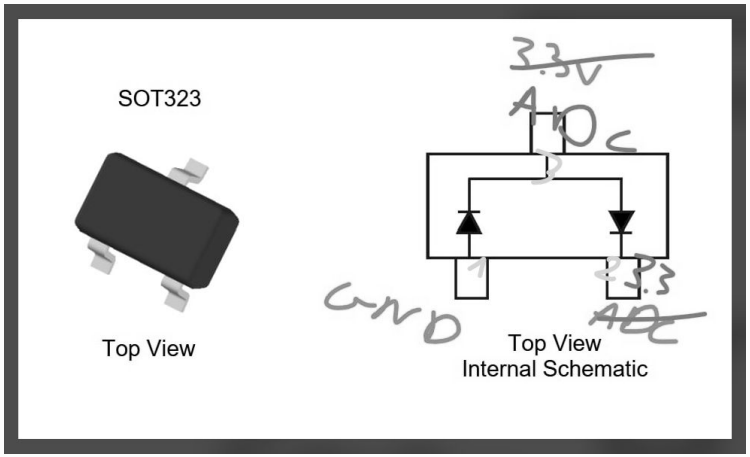




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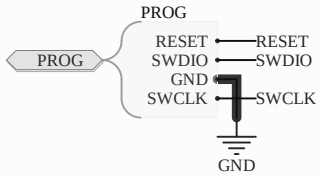
Protection diode - Vbat readout

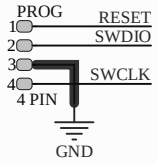




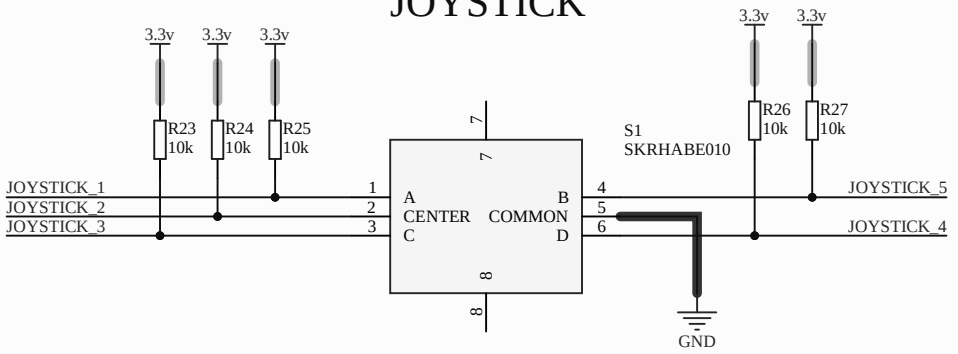
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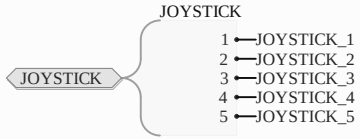
PROG





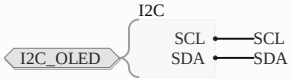
JOYSTICK

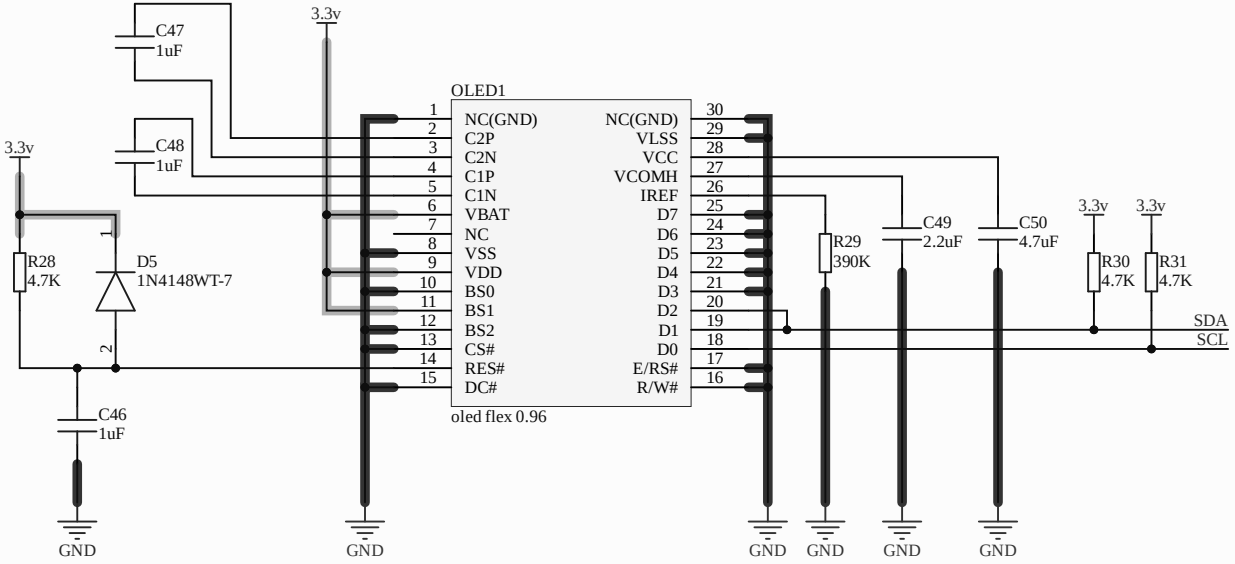


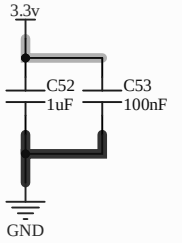


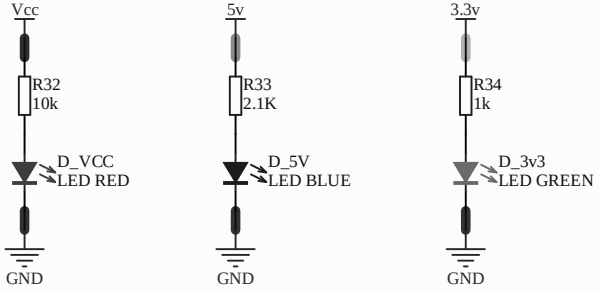
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OLED (flex)



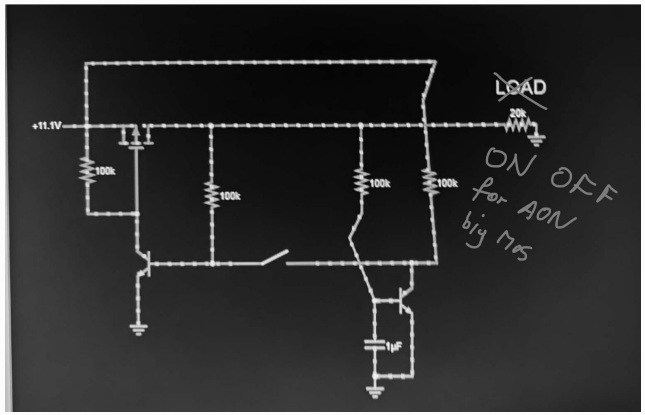


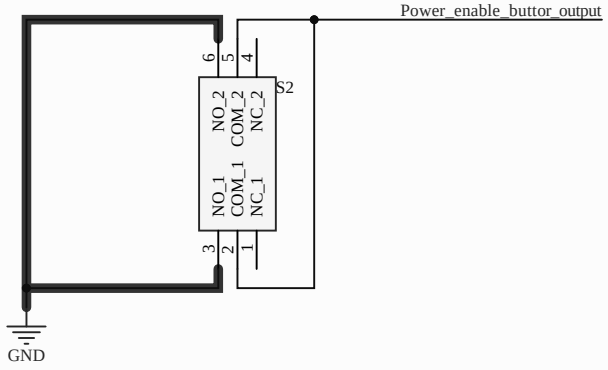




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SOFT LATCHING BUTTON!

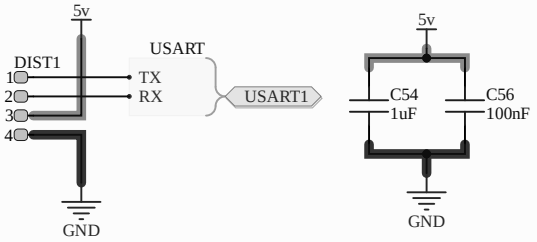


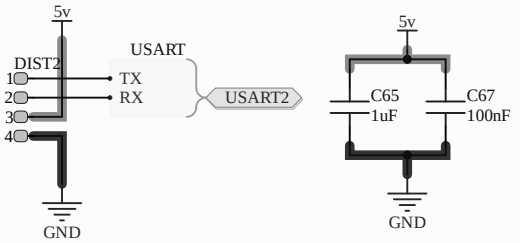


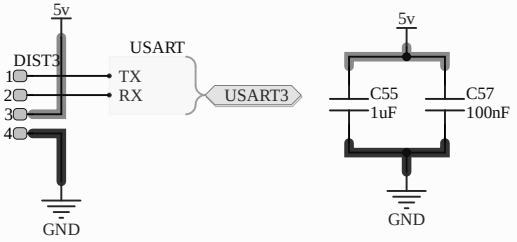
Power_enable_buttor_output PWR ENABLE

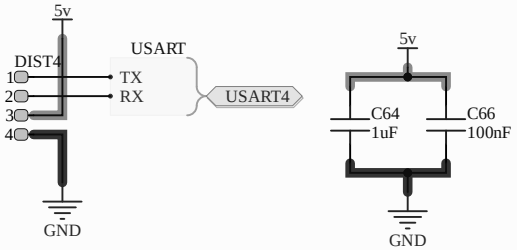
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LUNAs

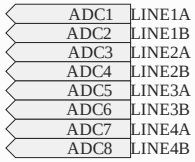


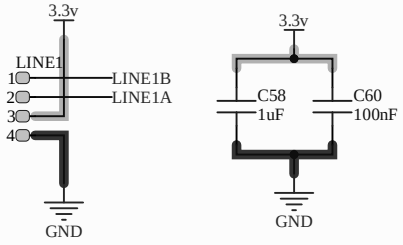


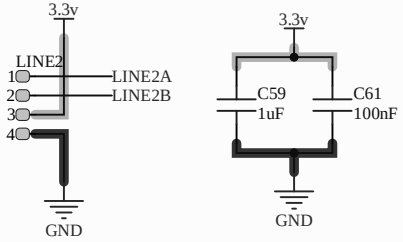


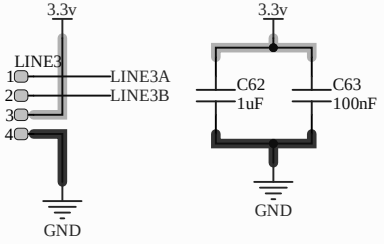


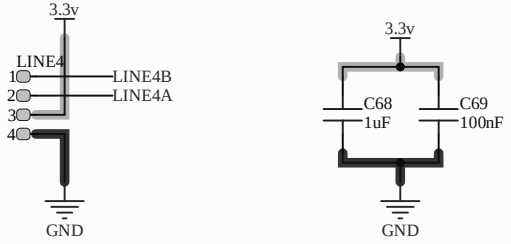
LINEs



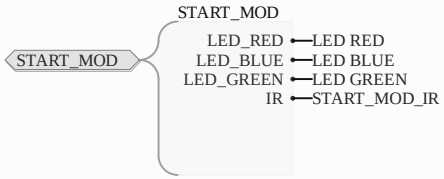






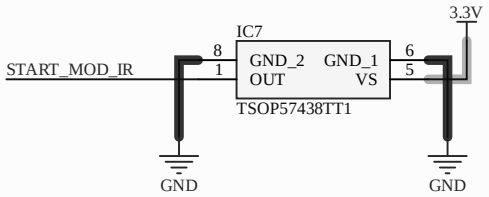


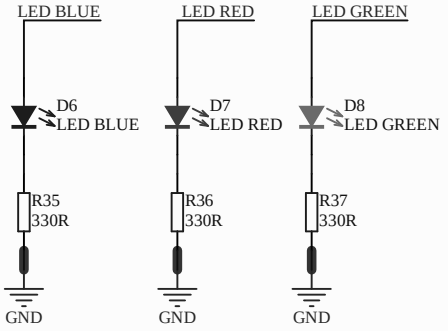
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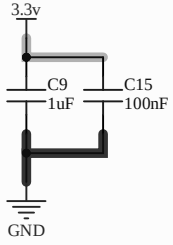


Start mod ulitilies

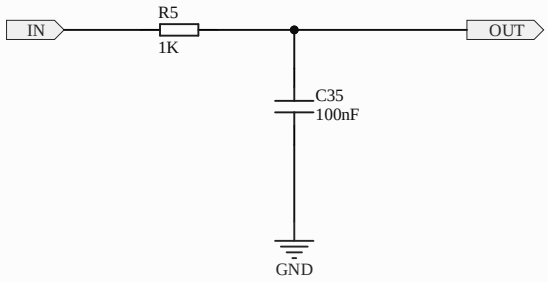
IR Reciever on board







Title		
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Title		
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